# SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY

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## “ASSIGNMENT”

Submitted by:

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Subject:

Fundamentales of AI and ML

Subject code:

24SBT110

Submitted to:

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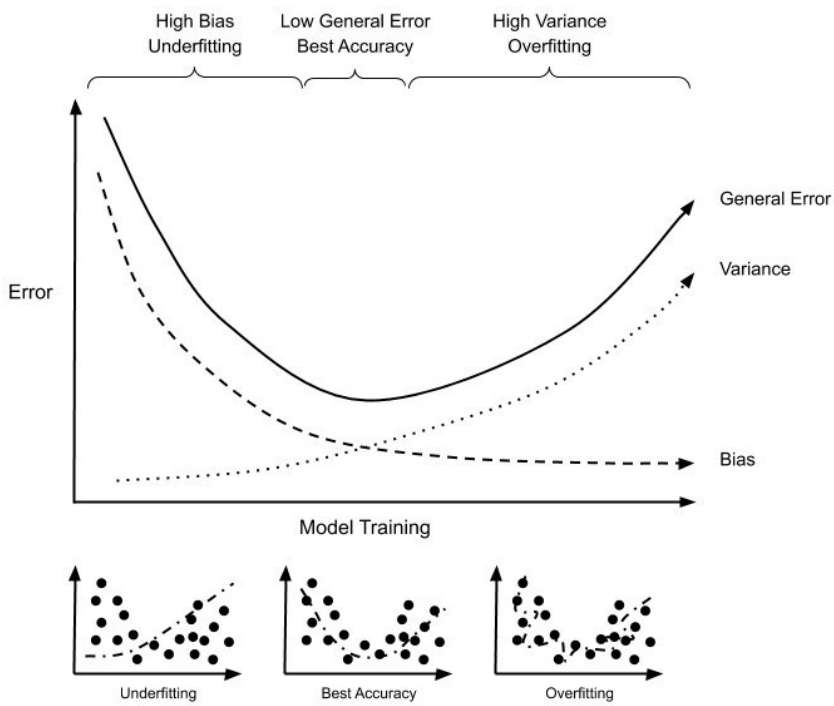
Academic Year 2026

1.Variance and Bias (Diagram, overfit, underfit)

- For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, low bias or high variance

Bias, Variance, Overfitting & Underfitting




Introduction

- Machine learning models aim to generalize well to unseen data.
- Two main sources of error: Bias and Variance.
- Understanding these helps to choose or design the best-fit model.

. Bias

- Bias is the error from oversimplifying the model.

- High bias → underfitting → model cannot capture patterns.
- Low bias → good pattern learning.
- Example: A straight line trying to fit curved data will perform badly.
- Effect: Both training and testing errors are high when bias is high.

## Variance

- Variance is the error from model being too sensitive to training data.
- High variance → overfitting → model learns noise along with patterns.
- Low variance → stable predictions across different datasets.
- Example: A very complex curve fitting every data point in a small dataset.
- Effect: Low training error but high testing error

## Underfitting vs Overfitting vs Best Fit

| Model Type | Bias | Variance | What Happens | Result |
|---|---|---|---|---|
| Underfit | High | Low | Too simple | Poor performance |
| Overfit | Low | High | Too complex | Good training, bad test |
| Best Fit | Low | Low | Just right | Good generalization |

- Underfitting: Model does not learn enough.
- Overfitting: Model learns noise and patterns.
- Best Fit: Balanced bias and variance → lowest error.

## Bias-Variance Trade-off

- Total error = $\text{Bias}^2$ + Variance + Irreducible Error.

- Increasing model complexity: o Bias decreases o Variance increases
- Decreasing complexity: o Variance decreases o Bias increases
- Goal: Minimize total error → find the best-fit model.

For the best-fit model, should we have:

- Low bias AND low variance

Explanation :

- Low bias → model captures true patterns.
- Low variance → model generalizes well and avoids overfitting.
- Achieves the best performance on unseen data.

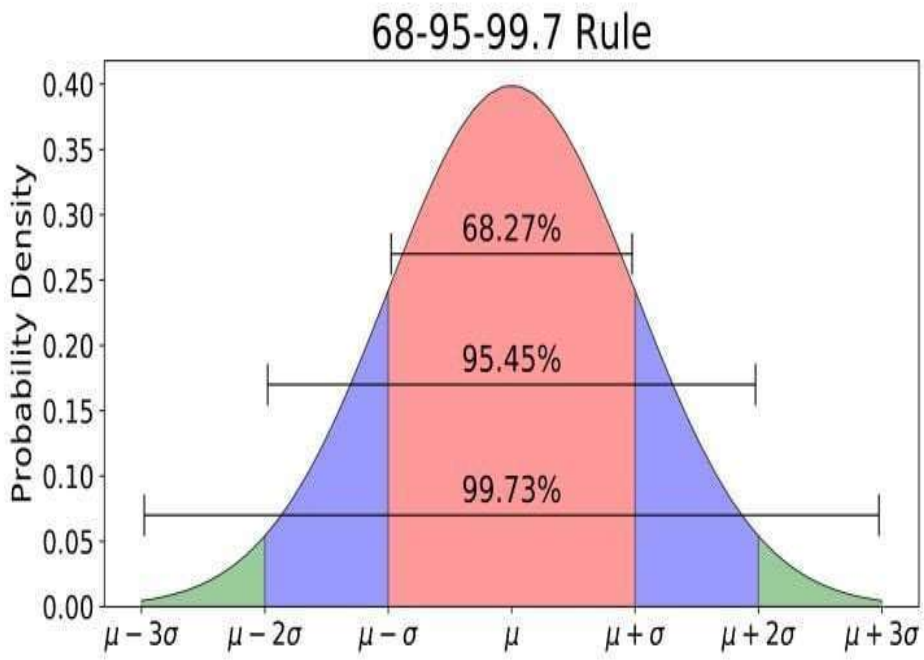
2. Take one Domain and draw the graph (Normal distribution) (Empirical rule)

Domain – Student Exam Scores

Domain: Marks scored by students in a class (0–100 marks)

- Assume the scores are normally distributed.
- Mean (μ): 70
- Standard Deviation (σ): 10

Normal Distribution Graph




## Empirical Rule (68-95-99.7 Rule)

- About 68% of students' scores lie within 1σ of the mean: 60–80 marks
- About 95% lie within 2σ of the mean: 50–90 marks
- About 99.7% lie within 3σ of the mean: 40–100 marks

- Normal distribution is bell-shaped and symmetric around the mean.
- Most data points fall close to the mean, fewer as you move away.
- The Empirical Rule helps in predicting the probability of values in the distribution.

## How to Label Your Graph for Assignment

- X-axis: Student marks
- Y-axis: Probability density / frequency
- Mark mean (μ = 70) at the center
- Shade areas for 1σ, 2σ, 3σ ranges with 68%, 95%, 99.7% labels

3. If $\mu = 55$, $\sigma_{4a} = 4$, $\sigma_{4\beta} = 10$, $\sigma_4 = c\ 15$, In this which is better

Problem

- Mean: $\mu = 55$
- Standard deviations: o $\sigma_{4a} = 4$ o $\sigma_{4\beta} = 10$ o $\sigma_4 c = 15$

Which is "better"?

Step 1: Understanding σ

- Standard deviation (σ) measures the spread of data around the mean.
- Smaller σ → data is tightly clustered around the mean → more consistent / predictable.
- Larger σ → data is spread out → more variability / less consistent.

Example:

- σ = 4 → most scores are close to 55
- σ = 10 → scores are more spread
- σ = 15 → scores vary a lot

Step 2: Use Empirical Rule (68-95-99.7%)

For μ = 55:

| σ | 68% range (±1σ) | 95% range (±2σ) | 99.7% range (±3σ) |
|---|---|---|---|
| 4 | 51–59 | 47–63 | 43–67 |
| 10 | 45–65 | 35–75 | 25–85 |
| 15 | 40–70 | 25–85 | 10–100 |

Observation:

- Smaller σ → narrower range, scores are more predictable.
- Larger σ → wider range, scores are more spread out → less reliable.

Step 3: Which is "Better"?

Better means more consistent and less variability.

- $\sigma_{4a} = 4 \rightarrow$ best, scores are tightly clustered around 55
- $\sigma_{4\beta} = 10 \rightarrow$ moderate
- $\sigma_{4c} = 15 \rightarrow$ worst, very spread out

Answer: $\sigma_{4a} = 4$ is the best distribution

Step 4: Graphical Representation (Optional for Assignment)

Draw three bell curves with the same mean $\mu = 55$:

- $\sigma_{4a}$ = narrowest curve
- $\sigma_{4\beta}$ = medium curve
- $\sigma_{4c}$ = widest curve

- This visually shows that smaller $\sigma$ = more concentrated / better.

4. Take one domain and build business Understanding

E-Commerce Customer Purchases

Domain: Online retail store – customer purchase data

1. Business Objective

- Understand customer purchasing behavior to:
  - Increase sales
  - Improve customer retention
  - Optimize marketing campaigns
- Example questions:
  - Which products are most popular?
  - Which customers are likely to make repeat purchases?
  - What is the average order value?

## Key Metrics / KPIs

- Revenue per customer
- Frequency of purchase
- Average order value
- Customer segmentation (high-value vs low-value customers)

## Business Understanding Steps

1. Identify the problem:
    - Increase sales by targeting high-value customers.
2. Determine data requirements:
    - Customer profile: ID, age, gender, location
    - Purchase history: date, product, quantity,

    price o Marketing interactions: emails,

    promotions
3. Define success criteria:
    - Increase repeat purchase rate by 20%
    - Increase average revenue per

    customer
4. Set AI/ML goal:
    - Predict which customers are likely to buy

    again o Segment customers based on purchase

    behavior

## How AI Helps in This Domain

- Predictive Analytics:
    - AI models (e.g., logistic regression, decision trees, random forest) can predict which customers are likely to make repeat purchases.
- Customer Segmentation:

    - AI clustering algorithms (e.g., K-Means, DBSCAN) can group customers into high-value, medium-value, and low-value segments.
- Personalized Recommendations:
    - AI-based recommendation systems suggest products to customers based on past purchases and preferences, increasing cross-selling and upselling.
- Churn Prediction:
    - AI detects customers at risk of leaving and helps in creating targeted retention campaigns.
- Marketing Optimization:
    - AI helps in deciding the best offer or promotion for each customer, maximizing ROI on campaigns.

## Example Output / Insights Expected

- Segmentation: High-value, medium-value, low-value customers  Predicted repeat buyers: Identify who is likely to purchase again
- Recommendations:
    - Personalized promotions to high-value customers o Discounts for medium-value customers to increase engagement o Reduce marketing spend on low-value customers